

Sculpting with Equations

How to build a coffee mug, a die, and a two-holed doughnut out of a single formula

Carlos Valero Valdés

Departamento de Matemáticas, Universidad de Guanajuato, México
valerocar@gmail.com

Abstract

Every shape in this article — a two-holed doughnut, a coffee mug with a handle, a rounded die with all twenty-one pips — is a *single equation*. No polygon meshes were sculpted, no 3D modeling software was harmed. The whole trick rests on two ideas simple enough to explain over coffee: you can do *arithmetic* with yes/no questions, and you can *soften* a yes/no question into a “maybe.” A third idea, warping space itself, then twists these creations like taffy. Everything is illustrated with free, open-source Python code,¹ so you can sculpt along at home.

1 A shape is a question

There are two very different ways to tell someone what a circle is. One is to give directions for *drawing* it: put your compass point here, open it this wide, spin. The other is to give a *test* for membership: the circle of radius 1 consists of exactly those points (x, y) for which

$$x^2 + y^2 = 1.$$

The first description is a recipe for a pencil. The second is a *question*: hand me any point in the plane and I will answer, using the formula, whether it lies on the circle. Shapes described this way are called *implicit*, and they are a quietly powerful idea: the formula knows about *every* point at once. Even better, if we relax the question from “are you *on* the circle?” to

$$x^2 + y^2 \leq 1 \quad (\text{“are you inside?”})$$

we get a *solid* disk instead of a hairline curve. Solid objects in 3D work the same way: $x^2 + y^2 + z^2 \leq 1$ is a solid ball, $x^2 + y^2 \leq 1$ is an infinite solid cylinder around the z -axis, and $z \geq 0$ is the half of space above the floor.

That is a fine start, but nobody drinks coffee out of a half-space. Real objects — mugs, dice, pretzels — look hopelessly beyond the reach of one formula. The point of this article is that they are not. We will build them from spheres, cylinders and planes the way a sculptor builds from clay: joining pieces, carving pieces away, and smoothing the seams with a thumb. Our thumb will be a parameter called k .

¹<https://github.com/valerocar/geometry-blender>

2 Arithmetic with yes and no

Mathematics has a wonderfully blunt device for turning a shape into a function. Given a set A (say, the solid disk), its *indicator function* χ_A answers the membership question with a number:

$$\chi_A(p) = \begin{cases} 1 & \text{if the point } p \text{ belongs to } A, \\ 0 & \text{if it does not.} \end{cases}$$

One for yes, zero for no. The surprise is that the standard ways of combining *sets* become ordinary *arithmetic* on these answers.

Take two sets A and B . A point belongs to the intersection $A \cap B$ when both answers are 1, and multiplication detects exactly that:

$$\chi_{A \cap B} = \chi_A \chi_B,$$

since a product of zeros and ones equals 1 only when every factor is 1. Multiplication is the word *and*. Negation is just as neat: the complement of A (everything *not* in A) has indicator

$$\chi_{A^c} = 1 - \chi_A,$$

which flips $0 \leftrightarrow 1$. Subtraction is the word *not*. And what about *or* — the union $A \cup B$? A first guess, $\chi_A + \chi_B$, almost works, but it pays double in the overlap where both answers are 1. The correction is to refund the overlap:

$$\chi_{A \cup B} = \chi_A + \chi_B - \chi_A \chi_B. \quad (1)$$

Finally, carving B out of A (the difference $A - B$) means “in A *and not* in B ,” and the dictionary translates it mechanically:

$$\chi_{A-B} = \chi_A (1 - \chi_B). \quad (2)$$

Where you have met this before

Formula (1) should ring a bell if you have studied probability: it is the shadow of the inclusion–exclusion rule $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. And the whole dictionary — *and* is $\times$, *not* is $1 -$ — is precisely how George Boole arithmetized logic in the 1850s. Blending shapes and blending truth values are, at bottom, the same trade.

One more ingredient connects this to the implicit shapes of Section 1. Our primitive solids all have the form “the region where some smooth function f is positive”:

$$P_f = \{p : f(p) \geq 0\}, \quad \text{e.g. } f(x, y, z) = 1 - x^2 - y^2 - z^2 \text{ for the ball.}$$

To get the indicator of P_f we simply post-process f with the *Heaviside step function* H , which crushes every positive number to 1 and every negative number to 0:

$$\chi_{P_f} = H \circ f, \quad H(s) = \begin{cases} 1 & s \geq 0, \\ 0 & s < 0. \end{cases}$$

Now we hold a complete toolkit. Want the region inside the ball *and* above the floor? Multiply their two indicators. Want the union of two cylinders? Use formula (1). Any finite sentence made of *and*, *or*, *not* applied to primitive shapes becomes one explicit formula. Engineers know this game as *constructive solid geometry*, and it is how CAD programs think about machine parts.

There is only one problem, and it is the interesting one: everything so far has *sharp edges*.

3 The softness knob

The Heaviside function is a cliff. It jumps from 0 to 1 with no warning, and every shape assembled from it inherits creases, corners, and seams. Sometimes that is what you want (a machined bracket); often it is not (a living form, a worn pebble, a doughnut). Calculus also despises jumps: you cannot differentiate a cliff.

The remedy is to replace the cliff by a ramp. Define, for each $k > 0$, the *sigmoid*

$$H_k(s) = \frac{1}{1 + e^{-s/k}}.$$

Figure 1 shows the whole family. Each H_k climbs smoothly from 0 to 1; the parameter k decides how briskly. For large k the climb is leisurely; as $k \rightarrow 0$ the ramp steepens and the family closes in on the Heaviside cliff. Two features deserve a pin in them:

- every curve in the family passes through the same point $(0, \frac{1}{2})$, and
- H_k never actually equals 0 or 1 — it hedges, answering “0.93” or “0.02” instead of yes or no.

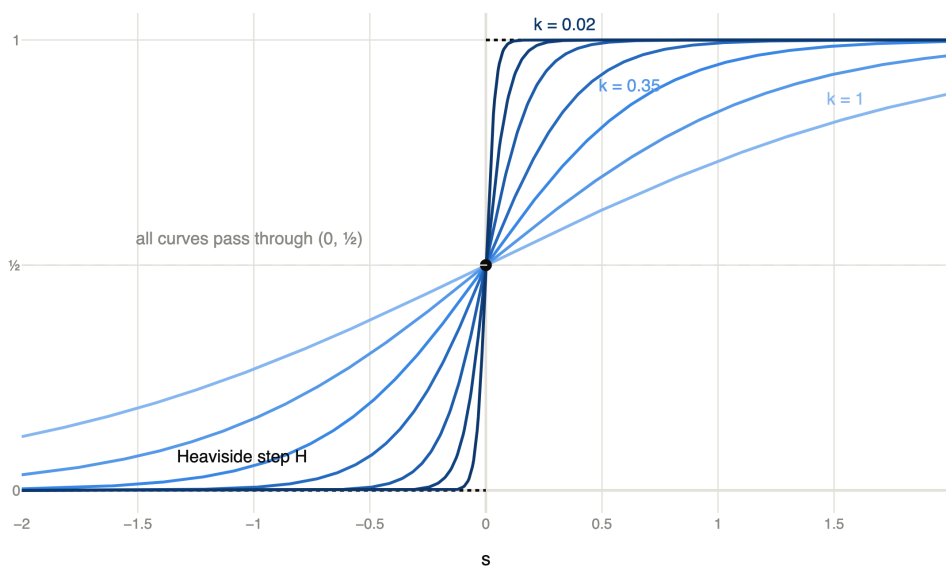
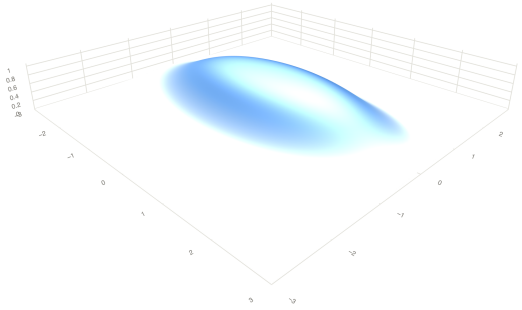


Figure 1: The sigmoid family $H_k(s) = 1/(1 + e^{-s/k})$ for values of k from 1 (lightest) down to 0.02 (darkest). As $k \rightarrow 0$ the sigmoids converge to the Heaviside step. All of them cross the value $\frac{1}{2}$ at $s = 0$ — the fact the whole method hangs on.

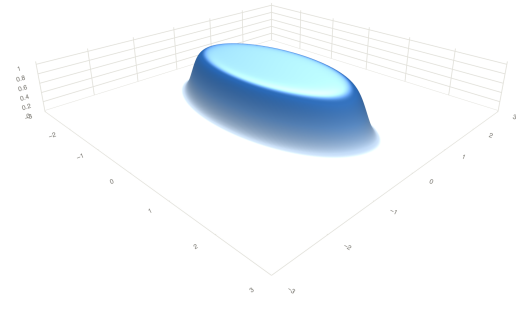
Now redo Section 2 with the cliff swapped for the ramp. For a shape P_f we define its *smooth indicator*

$$H_{f,k} = H_k \circ f,$$

a function that no longer answers the membership question with a curt yes/no but with a *score*: close to 1 deep inside the shape, close to 0 far outside, and gliding smoothly through $\frac{1}{2}$ as you cross the old boundary $f = 0$. Figure 2 shows the score function of an elliptical disk for two settings of k : each is a plateau of height 1 with softened shoulders, and the smaller k is, the more the plateau looks like a mesa with vertical cliffs.



(a) $k = 1/2$: gentle shoulders.



(b) $k = 1/10$: nearly a mesa.

Figure 2: The smooth indicator $H_{f,k}$ of the elliptical disk $f(x, y) = 1 - (x/2)^2 - y^2 \geq 0$, drawn as a landscape over the plane. Height is the membership score, from 0 (out) to 1 (in).

Here is the beautiful part. The blending formulas (1)–(2) never asked their inputs to be exact zeros and ones — they are plain arithmetic and accept scores just as happily. Feed them two smooth indicators $H_{f,k}$ and $H_{g,k}$ and they return smooth score-landscapes for the union, the intersection, and the difference of the two shapes. Figure 3 shows all three for a pair of overlapping disks: a two-humped massif, a slim central peak, and a crescent-shaped ridge. No creases anywhere — the arithmetic has melted them.

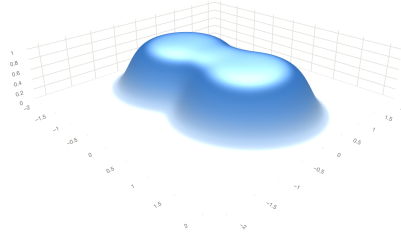
4 Cut at half height

A landscape of scores is lovely, but we were promised *shapes*. How do we climb back down? Remember the pinned fact: every sigmoid crosses $\frac{1}{2}$ exactly at $s = 0$. Consequently the score $H_{f,k}$ equals $\frac{1}{2}$ exactly where $f = 0$ — that is, exactly on the boundary of the original shape, whatever the value of k . So the rule is:

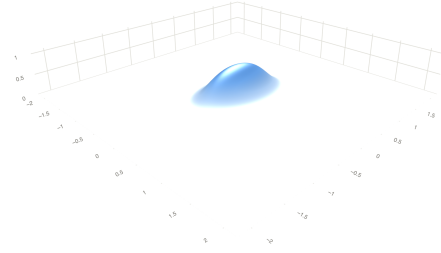
*slice the score landscape at height $\frac{1}{2}$,
and keep everything above the cut.*

For a single primitive this recovers the original circle, sphere, or cylinder without a hair out of place. For a *blend* it does something better: the half-height contour of, say, $H_f + H_g - H_f H_g$ is *not* the sharp-cornered outline of the set-union — it is a smoothed compromise, a curve that hugs the union at a distance and rounds off its crevices. That is a feature, not a bug: it is the mathematical version of running your thumb along the seam between two lumps of clay. Figure 4 shows the three half-height curves from the landscapes of Figure 3, with the original “ingredient” circles dashed underneath.

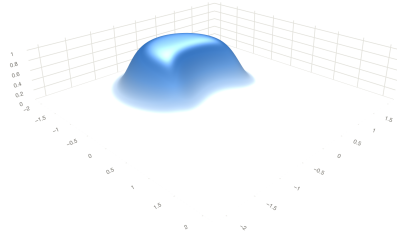
The knob k now has a direct sculptural meaning: it is the *radius of your thumb*. Figure 5 makes the point in 3D by uniting two balls three times, with three settings of k . Small k gives a crisp weld line; medium k melts a graceful neck between the balls; large k fuses them into a single pebble. Readers who remember lava lamps, or the liquid-metal villain of a certain 1991 film, have seen this aesthetic before: computer graphics people call such objects *metaballs* or *blobby models*, and they are blends of exactly this kind.



(a) union: $H_f + H_g - H_f H_g$



(b) intersection: $H_f H_g$



(c) difference: $H_f(1 - H_g)$

Figure 3: Score landscapes for two overlapping disks of radius 1 centered at $(\pm 3/4, 0)$, with $k = 1/4$. (Subscripts k dropped to reduce clutter.)

From here on, every picture in this article is obtained the same way: write down score functions for a few primitives, combine them with the arithmetic of Section 2, and draw the set of points where the final score equals $\frac{1}{2}$. One formula per object, as promised.

5 A double doughnut from scratch

Time to build something with real topology: a surface with *two* holes. The recipe has three lines.

Step 1: the body. Take two flattened ellipsoids, side by side and slightly overlapping — the regions where

$$e_{\pm}(x, y, z) = 1 - \left(\frac{x \mp 1}{1.15} \right)^2 - y^2 - \left(\frac{z}{0.55} \right)^2 \geq 0.$$

Blend their scores with the union formula; the result is a smooth, pillow-like slab with a waist (the blue body in Figure 6).

Step 2: the drills. Take two vertical cylinders of radius 0.42 through the centers of the ellipsoids:

$$c_{\pm}(x, y, z) = 0.42^2 - (x \mp 1)^2 - y^2 \geq 0,$$

again united into a single score function (the orange drills in the same figure).

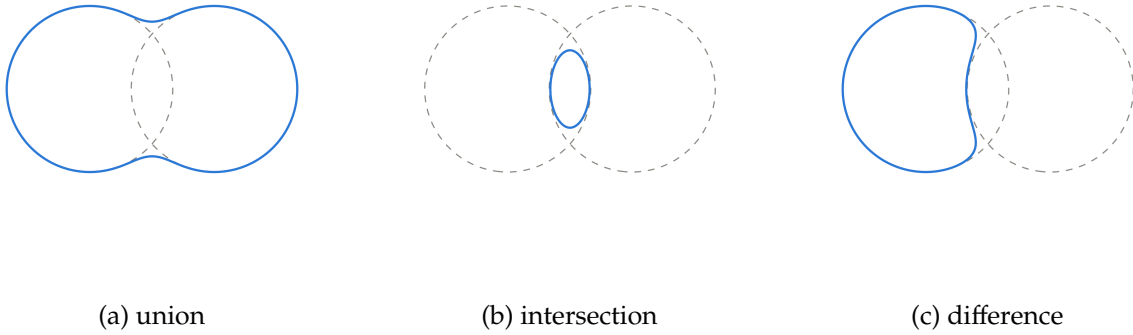


Figure 4: Cutting the landscapes of Figure 3 at height $\frac{1}{2}$ yields smooth blended curves (solid). The dashed circles are the two ingredients. Note how the union bulges to swallow the corner where the circles cross, and the difference scoops a rounded bite.

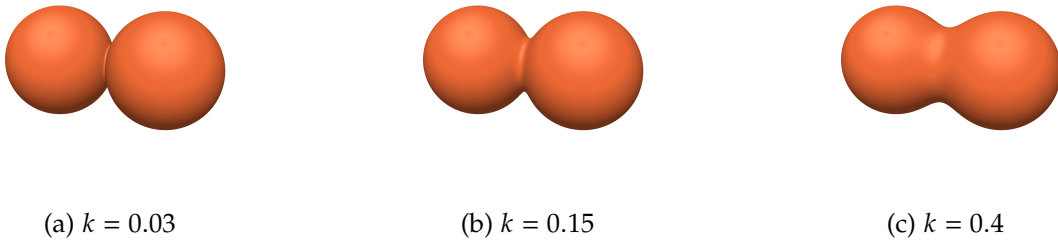


Figure 5: The union of two overlapping balls at three settings of the softness knob. The surfaces drawn are the points where the blended score equals $\frac{1}{2}$.

Step 3: drill. Subtract the second score from the first with formula (2) and slice at half height:

$$F(1 - G) = \frac{1}{2},$$

F = union score of the two ellipsoids,

 G = union score of the two cylinders.

The result (Figure 7) is the *double torus*, the mathematician's genus-two surface, with both tunnels smoothly flared as if the object had been cast in one piece. Written out in full, its equation would sprawl across half a page of exponentials — but we never need to see it, any more than a baker needs to see the chemical formula of a croissant. The recipe *is* the understanding.

6 The coffee-mug theorem

There is an old joke that a topologist is someone who cannot tell a coffee mug from a doughnut: each has exactly one hole (the handle, the hole), so each can be kneaded into the other without tearing. Let us take the joke literally and *manufacture* the mug.

The blank. Intersect an infinite vertical cylinder of radius 1 with the slab $-1 \leq z \leq 1$. Multiplying the two scores gives a solid puck.

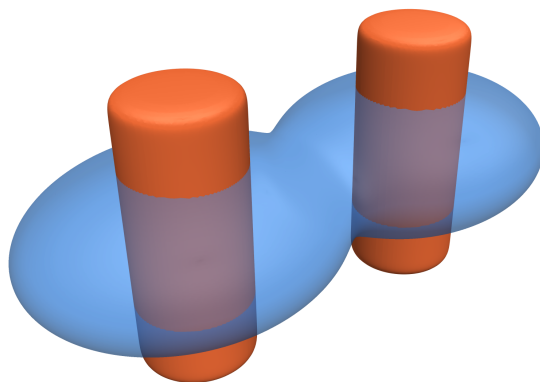


Figure 6: The ingredients: two blended ellipsoids (translucent blue) and two blended cylinders standing in for drill bits (orange).

The hollow. Take a thinner cylinder, radius 0.8, but cut it off below $z = -0.7$ (intersect with the half-space $z \geq -0.7$). Subtract it from the puck. What remains is a cup: walls, a floor, and nothing where the coffee goes.

The handle. Take a torus (a doughnut of tube radius 0.18), stand it upright, and press it into the side wall. The union formula welds it on, and because sigmoids never quite reach 0 or 1, the weld is a smooth fillet rather than a seam — exactly where a potter would smear the clay with a wet finger.

Figure 8 shows the result next to its doughnut twin. Count holes: the mug’s cavity does not count (a dent, however deep, can be pressed flat), but the handle’s loop does. One hole each. The joke checks out, and we built both punchlines out of sigmoids.

Genus, or how mathematicians count holes

The *genus* of a closed surface is the number of independent loops you can cut along without disconnecting it — informally, its number of “handles.” Sphere: genus 0. Torus and mug: genus 1. Our double doughnut: genus 2. A remarkable nineteenth-century theorem says that genus is the *only* thing that matters: two closed orientable surfaces can be deformed into one another exactly when their genus agrees. Everything else — size, dents, flourishes — is negotiable.

7 A die you could actually roll

Sharp objects are in reach too — and here the “flaw” of blending becomes its best feature. A cube is the intersection of six half-spaces: left of this wall, right of that one, below the ceiling, above

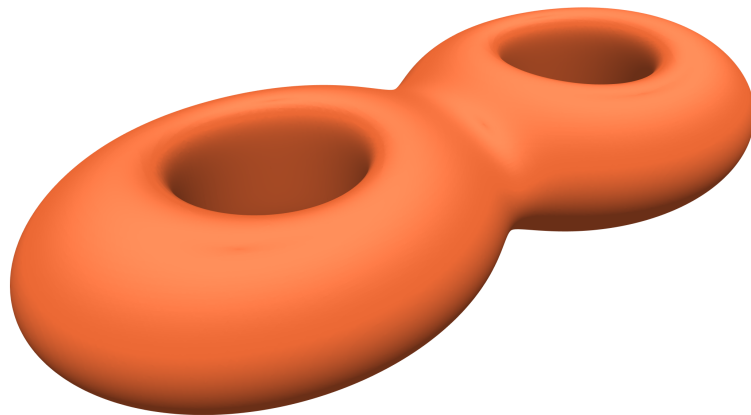


Figure 7: The finished double torus: ellipsoids united, cylinders subtracted, everything blended with $k = 0.18$. One equation.

the floor, and so on. Multiply the six scores. With a small k the result is practically a machinist's cube; with a moderate k (we used $k = 0.12$) the corners and edges come out *rounded*, because near an edge two scores are simultaneously hedging and their product dips below $\frac{1}{2}$ a little early. A manufacturer would pay extra for that: real dice are tumbled precisely to round their edges.

The pips are twenty-one little balls (that is $1 + 2 + \dots + 6$), each centered *on* a face so that half of it pokes into the cube, all subtracted in one long chain of formula (2). Each subtraction carves a neat hemispherical dimple. Figure 9 shows the result — the faces 1, 2 and 3 are visible, and opposite faces sum to seven, as regulation demands.

It is worth pausing on what we have *not* done. There is no list of vertices in this die, no mesh, no seams between “faces.” There is one smooth function of (x, y, z) , and the die is where that function exceeds one half. Roll it in your mind along any direction: every point of the surface answers to the same single formula.

8 Bending space itself

So far we have combined shapes while leaving space alone. The second technique of our title does the opposite: leave the shape's formula alone and *warp the space it lives in*.

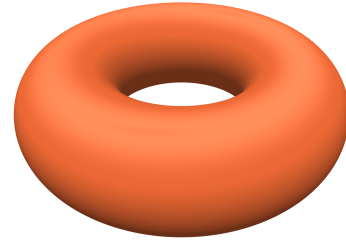
The principle fits in one line. Suppose a shape is carved out by the inequality $f(p) \geq \frac{1}{2}$, and suppose T is a transformation that moves points of space around. Then

$$\text{the warped shape is carved out by } f(T^{-1}(p)) \geq \frac{1}{2}.$$

Why the *inverse*? Because to decide whether a point p belongs to the warped object, we send p *back* to where it came from and ask the original formula there. (If a moving company shifts your



(a) (puck – hollow) $\cup$ handle



(b) its topological twin

Figure 8: The coffee-mug theorem, constructively. Both objects are single equations, and both have genus one: exactly one hole apiece.

house three meters east, your old floor plan still answers questions — provided you first walk three meters west while holding it.)

You have used special cases forever. Replacing x by $x - 3$ *moves* a graph three units right; replacing x by $x/2$ *stretches* it to double width. These are the *affine* transformations: translations, rotations, scalings, shears. They are loyal but predictable — straight lines stay straight.

The fun begins when the amount of transformation *depends on where you are*. Three classics, applied to a rounded bar in Figure 10:

- **Wave.** Replace x by $x - 0.35 \sin(2.2z)$: each horizontal slice slides sideways by an amount that oscillates with height. The bar sways like seaweed.
- **Taper.** Replace x, y by $x/s(z), y/s(z)$ with $s(z) = 1 - 0.28z$: each slice is scaled by a factor that shrinks with height. The bar becomes an obelisk.
- **Twist.** Rotate each slice by an angle proportional to its height — replace (x, y) by $(x \cos \alpha z + y \sin \alpha z, -x \sin \alpha z + y \cos \alpha z)$. The bar becomes barley sugar.

Note what the box above buys us: the warped objects are *still single equations* — we literally substitute the new expressions into the old formula. Blending and warping therefore compose freely, and this freedom is where the sculpting gets luxurious. As a finale, take an ordinary torus but flatten its circular cross-section into a thin blade; then let the blade's tilt angle *depend on the azimuth* — as you travel once around the ring, the cross-section rotates once about the ring's core. The result (Figure 11) is a twisted ribbon of a doughnut, the kind of object you would swear needed an artist and a 3D-modeling package. It is four lines of substitutions.

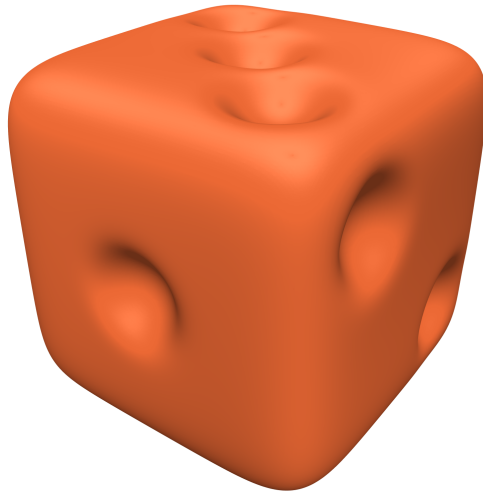
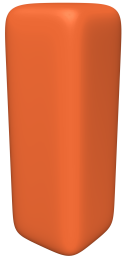


Figure 9: One equation, twenty-seven primitives: six half-spaces multiplied together ($k = 0.12$ rounds the edges for free) minus twenty-one pip-balls.

Where these ideas live in the wild

Blending indicator functions is the mathematical backbone of *implicit modeling* in computer graphics: James Blinn introduced “blobby” molecular surfaces in 1982, and modern signed-distance-field renderers on <https://www.shadertoy.com> sculpt entire worlds this way, in real time, inside your browser. The global bends, tapers and twists were catalogued for graphics by Alan Barr in 1984 and are now standard “deformers” in every animation package. The pictures in this article were extracted from their equations by *marching cubes* (Lorensen & Cline, 1987), the workhorse algorithm that also turns MRI scans into 3D anatomy.



(a) the bar



(b) waved



(c) tapered



(d) twisted

Figure 10: Global deformations of a blended bar (itself the intersection of six soft half-spaces). In each case the bar's formula is untouched; we only substitute new expressions for x, y, z before asking it the membership question.

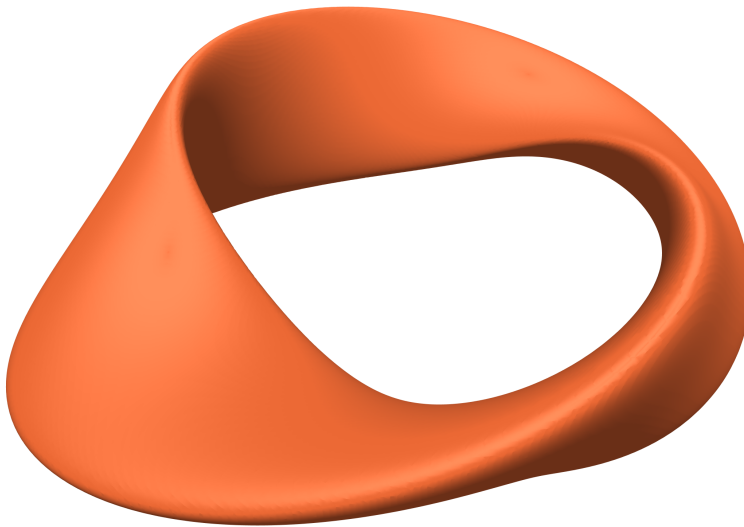


Figure 11: Finale: a torus with a flattened cross-section that rotates once as it travels around the ring — a warp whose rotation angle depends on the azimuth. Still one equation.

9 Sculpt your own

Every figure here was produced with a small open-source Python package, `gblend`,² which makes the blending arithmetic feel like set theory: the Python operators `&`, `|` and `-` are wired to the product, the inclusion–exclusion formula, and the carve-out formula of Section 2. The double torus of Figure 7, in its entirety:

```
from gblend.geometry import Ball3D, Cylinder3D

k = 0.18
e1 = Ball3D(k=k); e1.scale(1.15, 1, 0.55); e1.translate(-1, 0, 0)
e2 = Ball3D(k=k); e2.scale(1.15, 1, 0.55); e2.translate(+1, 0, 0)
c1 = Cylinder3D(x0=-1, r_val=0.42, k=k)
c2 = Cylinder3D(x0=+1, r_val=0.42, k=k)

double_torus = (e1 | e2) - (c1 | c2)    # blend, then drill
```

The mesh is extracted with `scikit-image`’s marching cubes and rendered with `plotly` — free tools, a laptop, no licenses.

Beyond the fun, there is a moral. We usually meet functions as graphs to be plotted and formulas as things to be solved. Here they played a different role: a formula was a *material* — something to be cast, carved, welded and twisted. The dictionary was small (multiply for *and*, flip for *not*, refund the overlap for *or*, soften the step, cut at one half, warp with a substitution), yet it was enough to make a mug, a die, and a two-holed doughnut out of thin air. Pick a shape on your desk and ask: what is *its* equation? You now own every tool needed to answer.

Further reading. J. Blinn, *A generalization of algebraic surface drawing*, ACM Transactions on Graphics 1 (1982) — the birth of blobby modeling. A. Barr, *Global and local deformations of solid primitives*, SIGGRAPH (1984) — bends, tapers and twists. W. Lorensen and H. Cline, *Marching cubes*, SIGGRAPH (1987) — from equations to triangles. A. Ricci, *A constructive geometry for computer graphics*, The Computer Journal 16 (1973) — blending unions and intersections before it was cool.

²<https://github.com/valerocar/geometry-blender> — install with `pip`, then run the article’s script `make_figures.py` to reproduce every image.